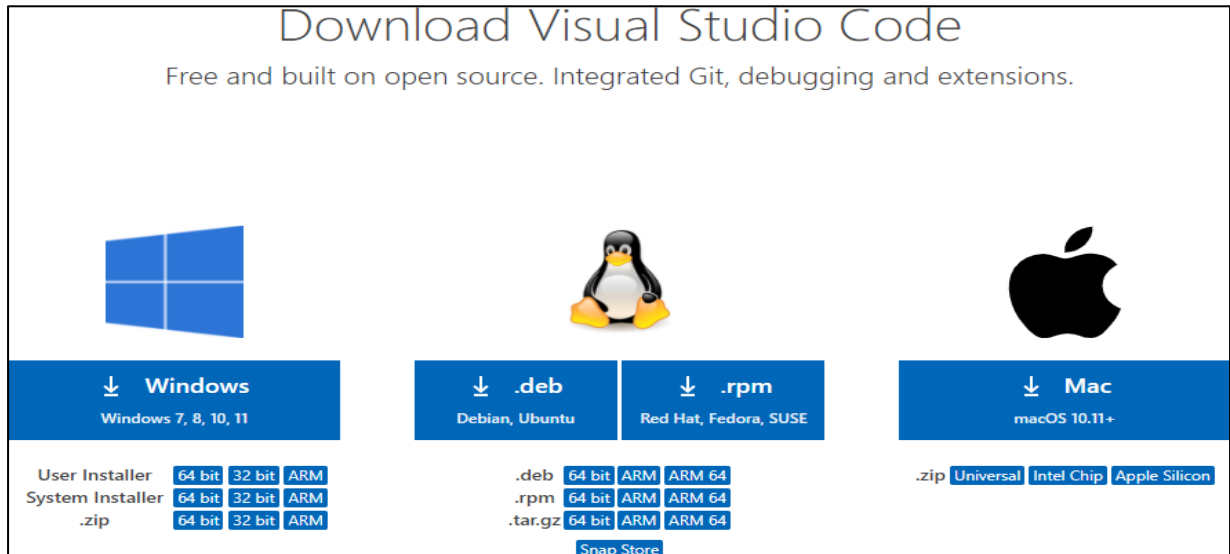


1.5 Installation & Setup of VS Code Editor

Step 1: Downloading VS Code (from official site)

Visit <https://code.visualstudio.com/> and download the installer.

Press the "**Download for Windows**" button on the website to start the download of the Visual Studio Code Application.



Step 2: Installation process (Windows/Linux/Mac)

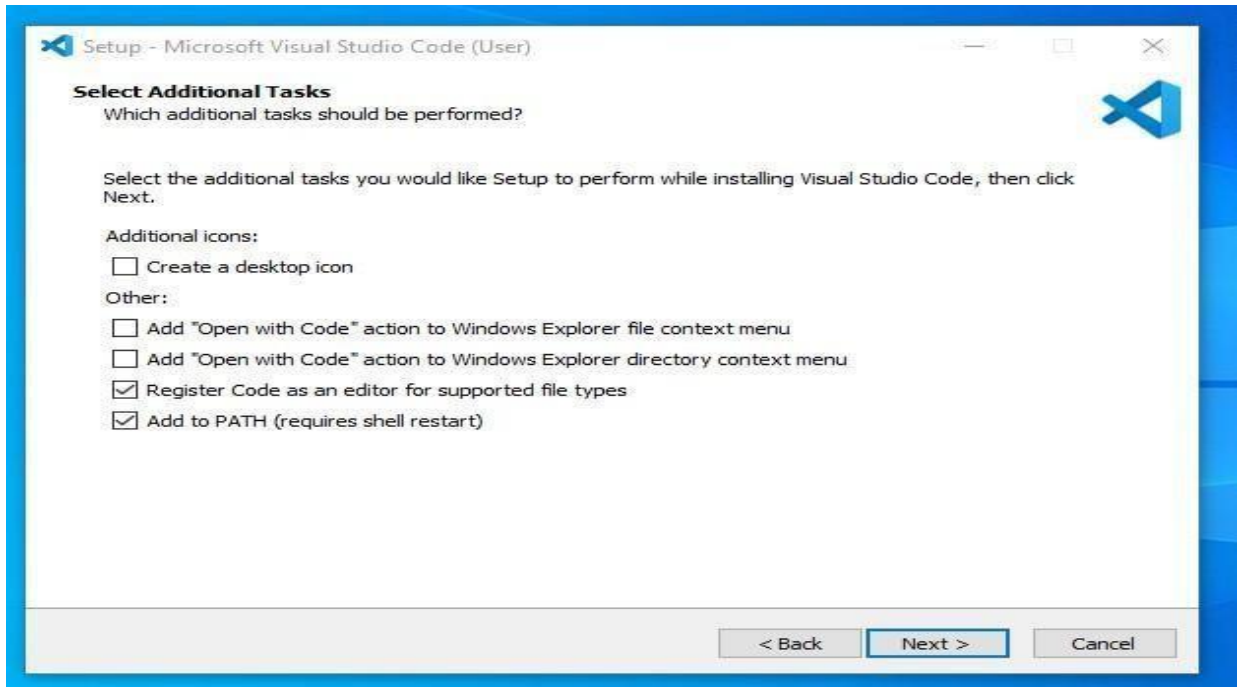
Run the installer → Accept License → Choose location → Install.

After the Installer opens, it will ask you to accept the terms and conditions of the Visual Studio Code. Click on **I accept the agreement** and then click the **Next** button.



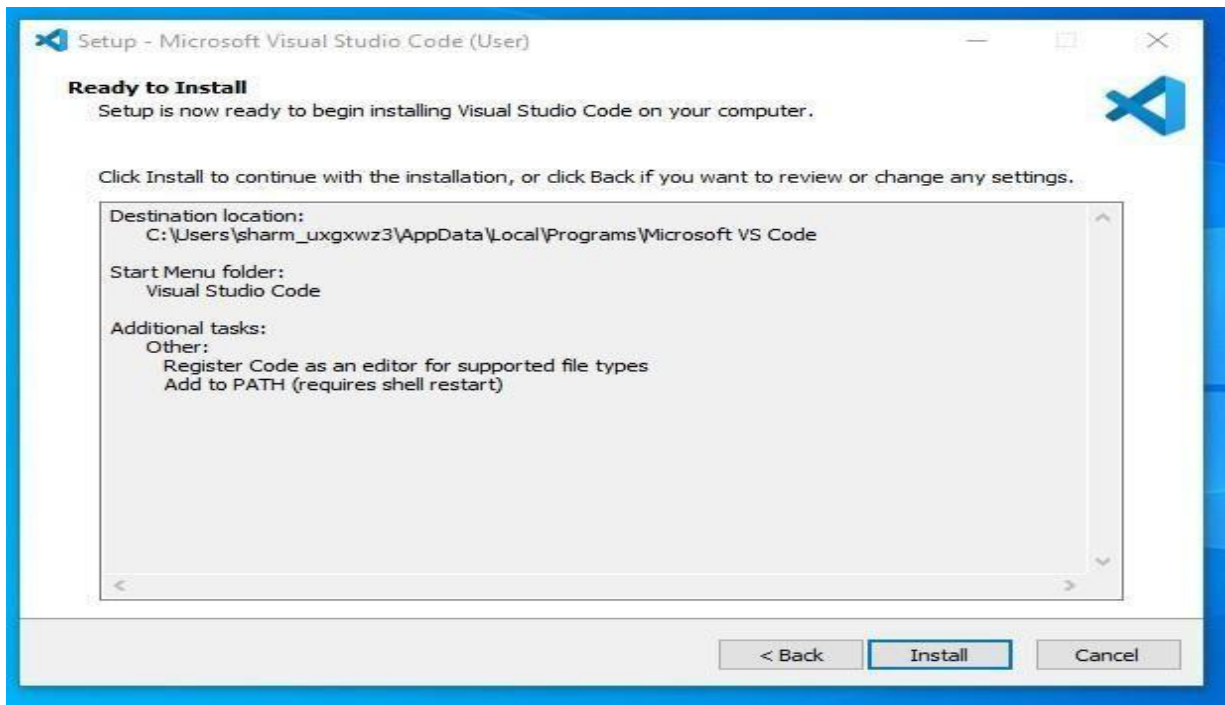
Step 3: Choose the location data for running the Visual Studio Code

Choose the location data for running the Visual Studio Code. It will then ask you to browse the location. Then click on the **Next** button.



Step 4: installation setup.

Then it will ask to begin the installation setup. Click on the **Install** button.



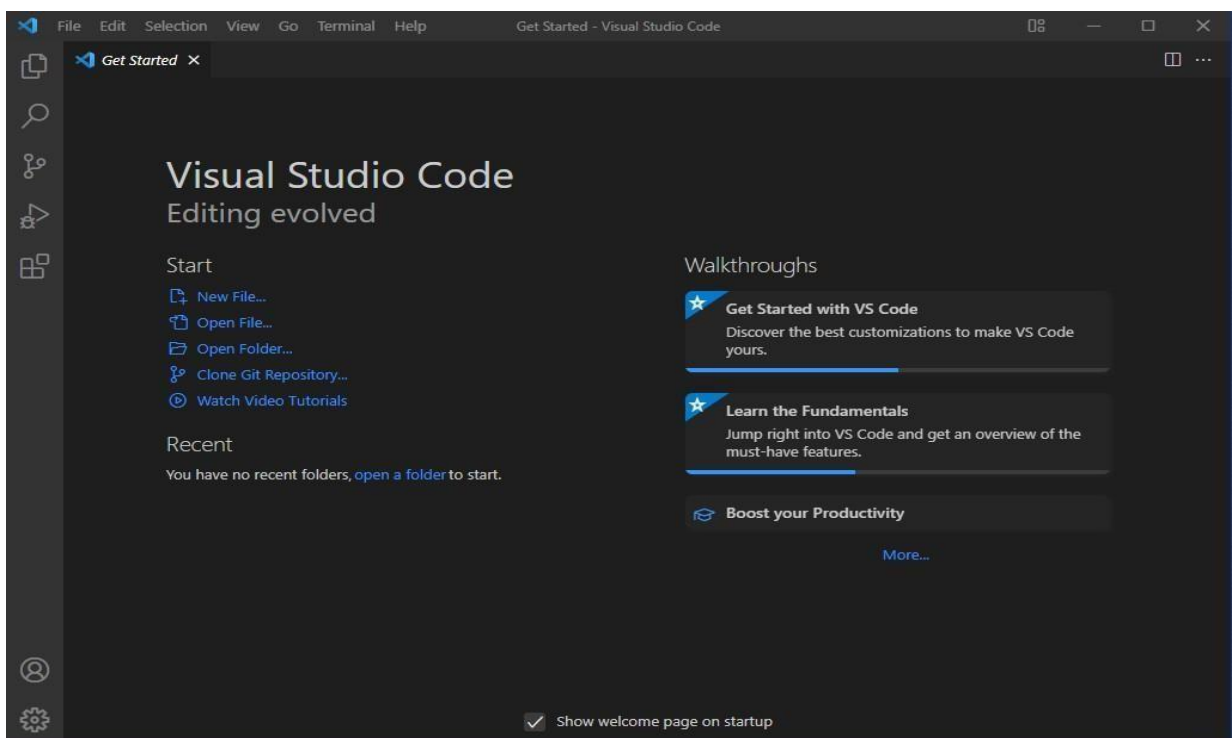
Step 5: Launch Visual Studio Code

After the Installation setup for Visual Studio Code is finished, it will show a window like this below. Tick the "**Launch Visual Studio Code**" checkbox and then click **Next**.



Step 6: Start Your Coding in Visual Studio

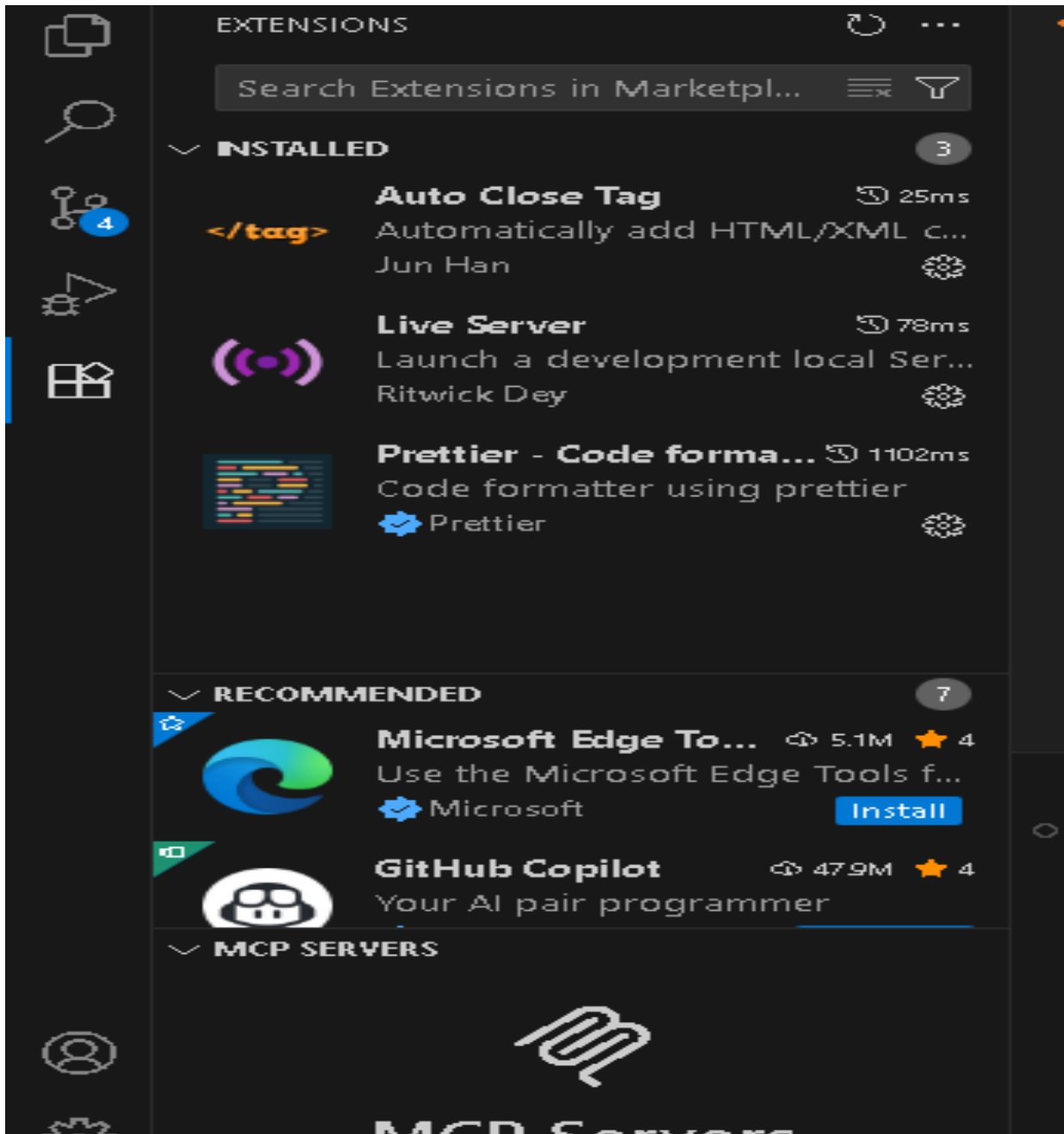
After the previous step, the Visual Studio Code window opens successfully. Now you can create a new file in the Visual Studio Code window and choose a language of yours to begin your programming journey!



1.5.1 How to Add Extensions in VS Code

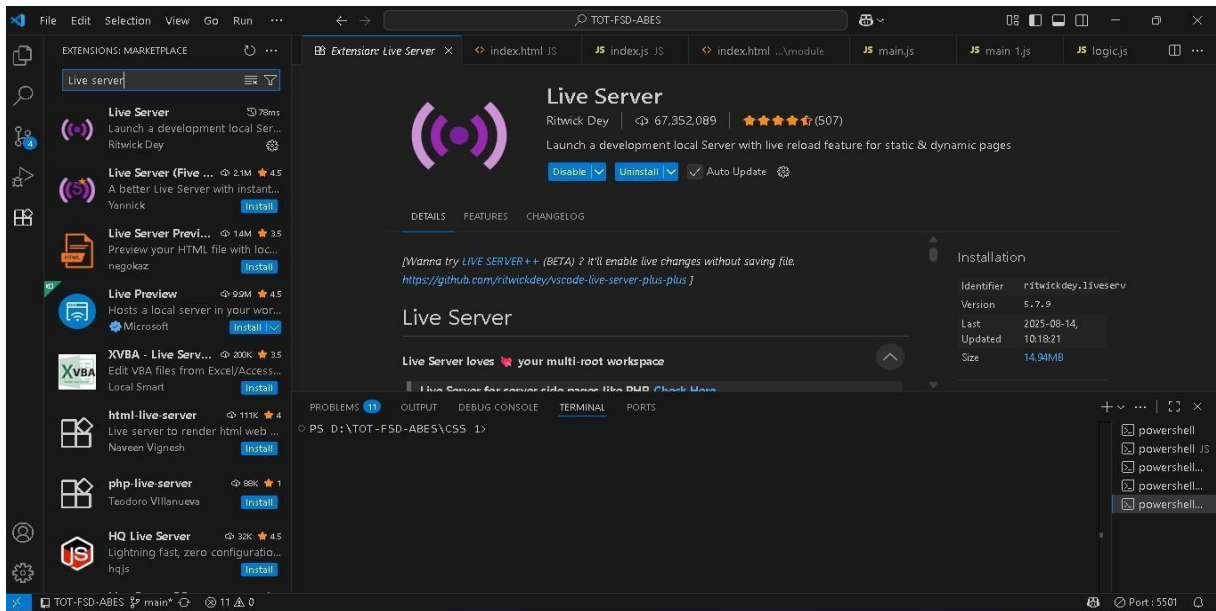
Step 1: Open Extensions Panel

- Open Visual Studio Code.
- On the left sidebar, click the Extensions icon (looks like 4 small squares).
- Shortcut: Ctrl + Shift + X



Step 2: Search for an Extension

- At the top of the Extensions panel, type the name of the extension you want.
- Example: Python, C/C++, Prettier - Code Formatter, Live Server



Step 3: Install the Extension

- From the search results, click the Install button next to the extension.
- It will download and install automatically.

Step 4: Reload if Needed

- Some extensions require a reload of VS Code.
- If prompted, click Reload.

Step 5: Manage Extensions

- Go to the Installed tab in the Extensions panel.
- From here you can:
 - Enable / Disable
 - Update
 - Uninstall

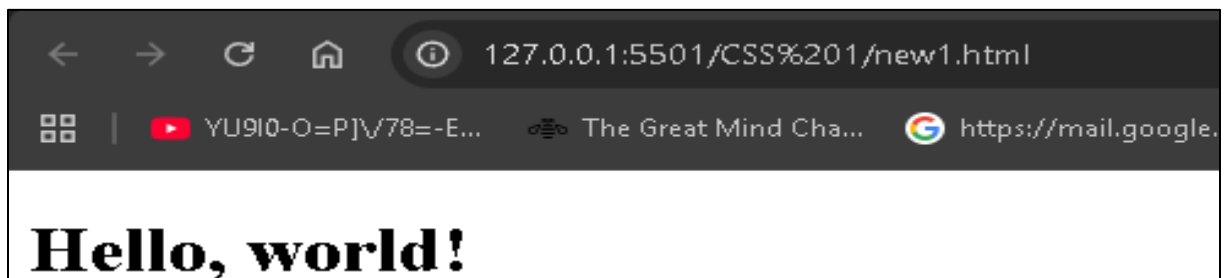
First sample HTML Program in VS Code:

- Open **Visual Studio Code**.
- Create a **new folder** (for example: MyFirstHTML).
- Inside the folder, create a new file and save it as:
index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Web Page</title>
</head>
<body>
  <h1>Hello, world!</h1>
</body>
</html>
```

Output

Output: Displays 'Hello, Web Technology!' in browser.



Second HTML Program in VS Code

- Open **Visual Studio Code**.
- Create a **new folder** (for example: MySecondHTML).
- Inside the folder, create a new file and save it as:
New.html

```
<!DOCTYPE html>
<html>
<head>
  <title>College Info</title>
</head>
<body>
  <h1>Jitendra Kumar Chauhan</h1>
  <h2>ABES Engineering College </h2>
  <h3>CSE-DS Department</h3>
</body>
</html>
```

ASSIGNMENT -1

1. Define Web Technologies. Explain different types of Web Technologies with suitable examples.
2. Difference between Front-End Developer and Back-End Developer.
3. Explain Client-Server Architecture with diagram
4. Create a simple HTML webpage in VS Code with the following details:
Instructions:
 1. Create a new folder named Web-Assignment on your computer.
 2. Inside the folder, create a file named index.html.
 3. The webpage must include the following:
 - A title in the browser tab: *“My first HTML Page”*.
 - A main heading (H1) containing your Name, Father’s Name, and Mother’s Name.
 - A subheading (H2) mentioning your Branch and Section.
 - A paragraph (P) that contains your address in 3–4 lines.
 4. Save the file and run it in a web browser to check the output.

1.6 Introduction of Git & GitHub

1.7 Installation & Configuration of Git Account

1.1 Introduction of Git & GitHub

What is a Version Control System (VCS)?

— Imagine you are writing your **Project Report**.

- Day 1: You save it as report.docx.
- Day 2: You add more content and save as report-final.docx.
- Day 3: You make changes and save as report-final2.docx.

Soon, your folder looks like this:

report.docx
report-final.docx
report-final2.docx
report-final2-new.docx

Confusing, right? Which one is the latest? Which one has the correct data? What if you want to **go back** to Day 1's version?

Version Control System (like Git) solves this problem!

- Git keeps track of every change (like a time machine).
- You can go back to *any previous version* easily.
- You don't need to create multiple final-final.docx files anymore!

Definition:

A **Version Control System (VCS)** is a tool that **records changes to files over time** so that you can recall specific versions later.

◆ What is Git?

Think of **Git as a CCTV Camera for your code**.

- It watches every change (who did it, when, what was changed).
- It stores snapshots (commits) of your project.
- You can rewind and see the history.
- Multiple developers can work together and Git will track “who changed what”.

◆ In technical words:

Git is a **distributed version control system**. It runs on your **local machine** and manages code versions, branches, and merging.

- **Git** = Version Control System (VCS).
- Tracks project changes → allows rollback.
- Enables collaboration among developers.

◆ What is GitHub?

If **Git** is your **CCTV camera at home**, then **GitHub** is your **cloud storage (like Google Drive)** where you upload those **CCTV recordings**.

- GitHub is an **online platform** where you can store your Git repositories.
- A **cloud-based platform** hosting Git repositories.
- It helps you **share code with teammates**, manage collaboration, and contribute to open-source.
- Provides features like:
 - **Pull requests** (suggesting changes)
 - **Issues** (tracking bugs/tasks)
 - **Forks** (copy of someone else's repo)
 - **Branches** (parallel development)
- Think of Git as **your notebook** and GitHub as **Google Drive for your notebook**.
- **Git vs GitHub – Relation & Difference**

Git	GitHub
Tool to track changes (VCS).	Platform to host Git repositories.
Works on your local system .	Works online (cloud) .
Can be used alone (without GitHub).	Cannot work without Git.
Example: Track project locally.	Example: Share project with your team worldwide.

- Git = **Notebook** (where you write your code & track changes).
- GitHub = **Library** (where you store your notebook so that others can read, borrow, or contribute).

Why Use Both Git & GitHub Together?

- **Git alone** → You can track changes locally, but **can't collaborate easily**.
- **GitHub alone** → It's just a website, can't track versions without Git.
- **Together** → You get **version control + cloud hosting + collaboration tools**.

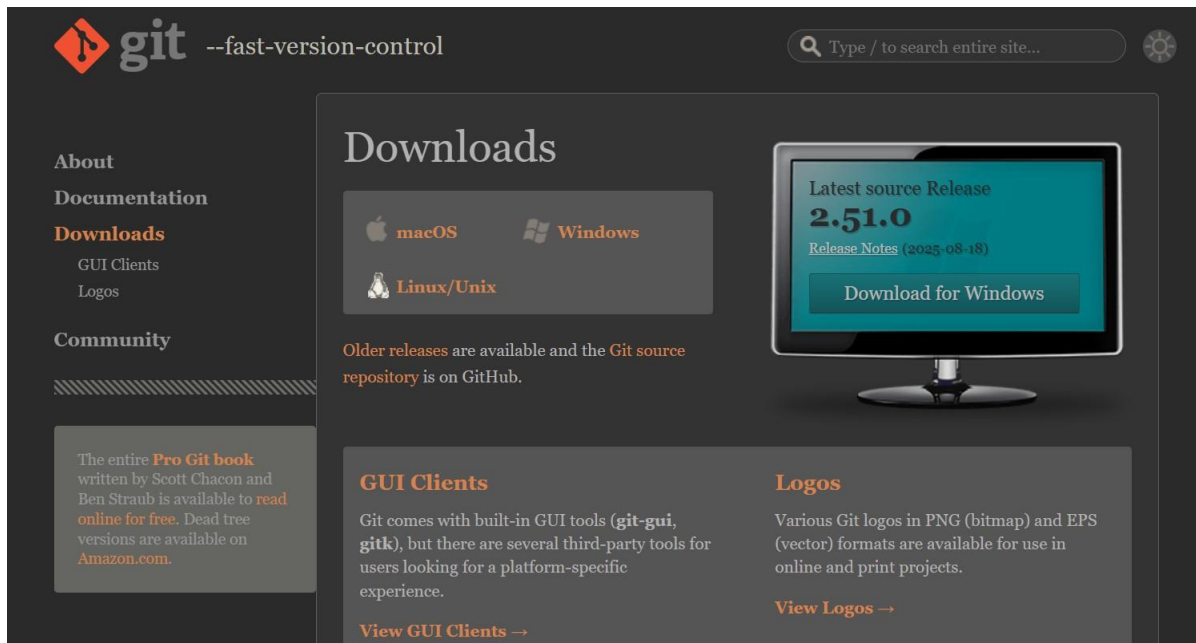
Example:

Imagine you and your friends are building a **mini web project**:

- On your laptop, Git tracks your changes (like saving checkpoints).
- You push it to GitHub → your friends can see and pull updates.
- They add new features → push again → you pull back.
- GitHub acts as a **shared code hub**, Git acts as the **history manager**.

1.2 Installation & Configuration of Git Accountsteps:

1. Download: <https://git-scm.com/downloads>



2. Install with default settings.
3. Open terminal (CMD/VS Code terminal) → check version:

git --version

```
Command Prompt
C:\Users\Suresh Verma>git --version
git version 2.46.0.windows.1
C:\Users\Suresh Verma>
```

Configure Git (first-time only):

```
git config --global user.name "Your Name"
git config --global user.email "your_email@example.com"
```

```
git config --list # check configuration
```

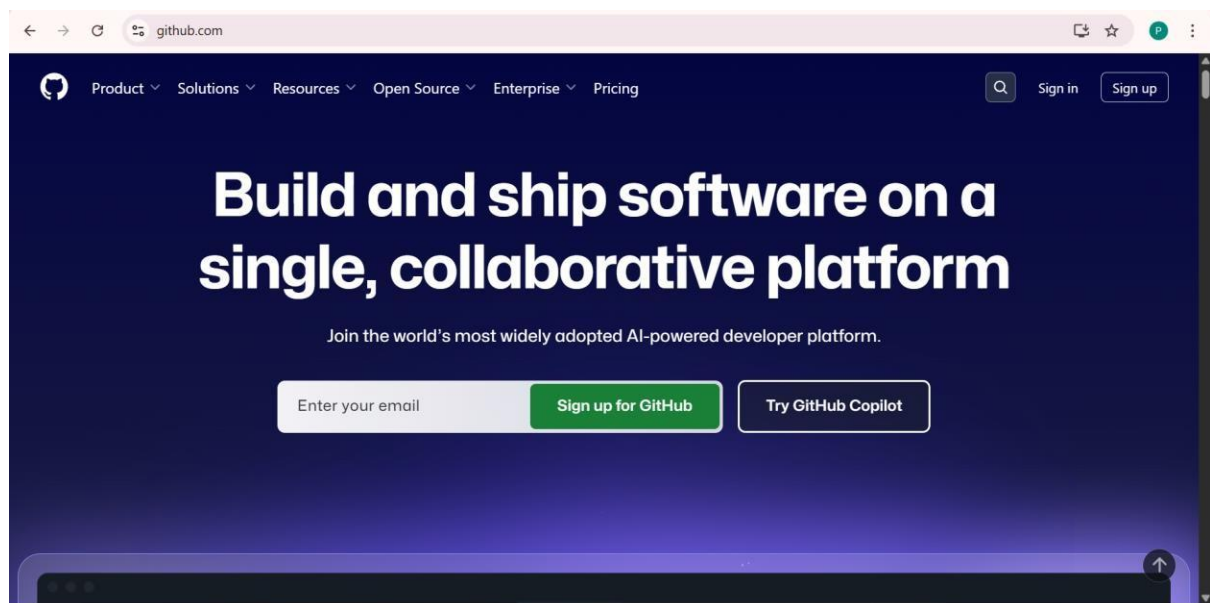
```
C:\Users\Suresh Verma>git config --global user.email "pranshi100verma@gmail.com"

C:\Users\Suresh Verma>git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=true
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com:https=httppath=true
init.defaultBranch=main
user.email=pranshi100verma@gmail.com
user.name=Pranshi Verma

C:\Users\Suresh Verma>
```

4. Creating a GitHub Account

1. Go to <https://github.com/>



2. Sign up with email → verify → choose username.

Sign up for GitHub · GitHub

github.com/signup?ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F&source=header-home

Already have an account? [Sign in](#)

Create your free account

Explore GitHub's core features for individuals and organizations.

[See what's included](#)

Sign up for GitHub

Continue with Google

or

Email

Password

Username

Your Country/Region

Email preferences

India

Receive occasional product updates and announcements

Create account >

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.

Sign up for GitHub · GitHub

github.com/signup?ref_cta=Sign+up&ref_loc=header+logged+out&ref_page=%2F&source=header-home

Already have an account? [Sign in](#)

Create your free account

Explore GitHub's core features for individuals and organizations.

[See what's included](#)

Sign up for GitHub

Continue with Google

or

Email

Password

Username

Your Country/Region

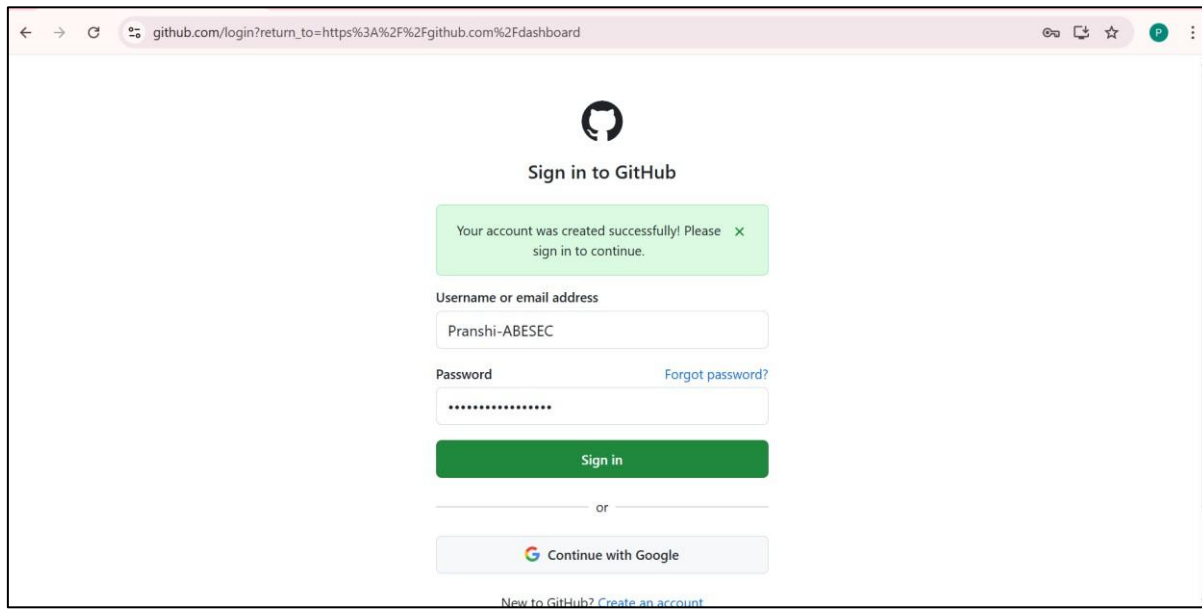
Email preferences

India

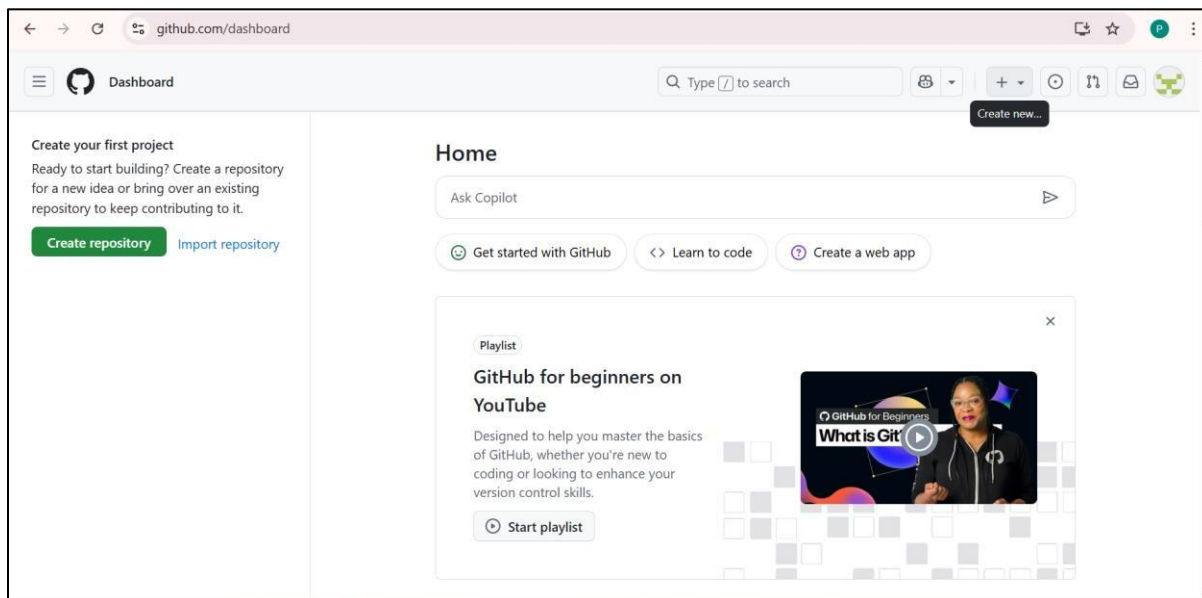
Receive occasional product updates and announcements

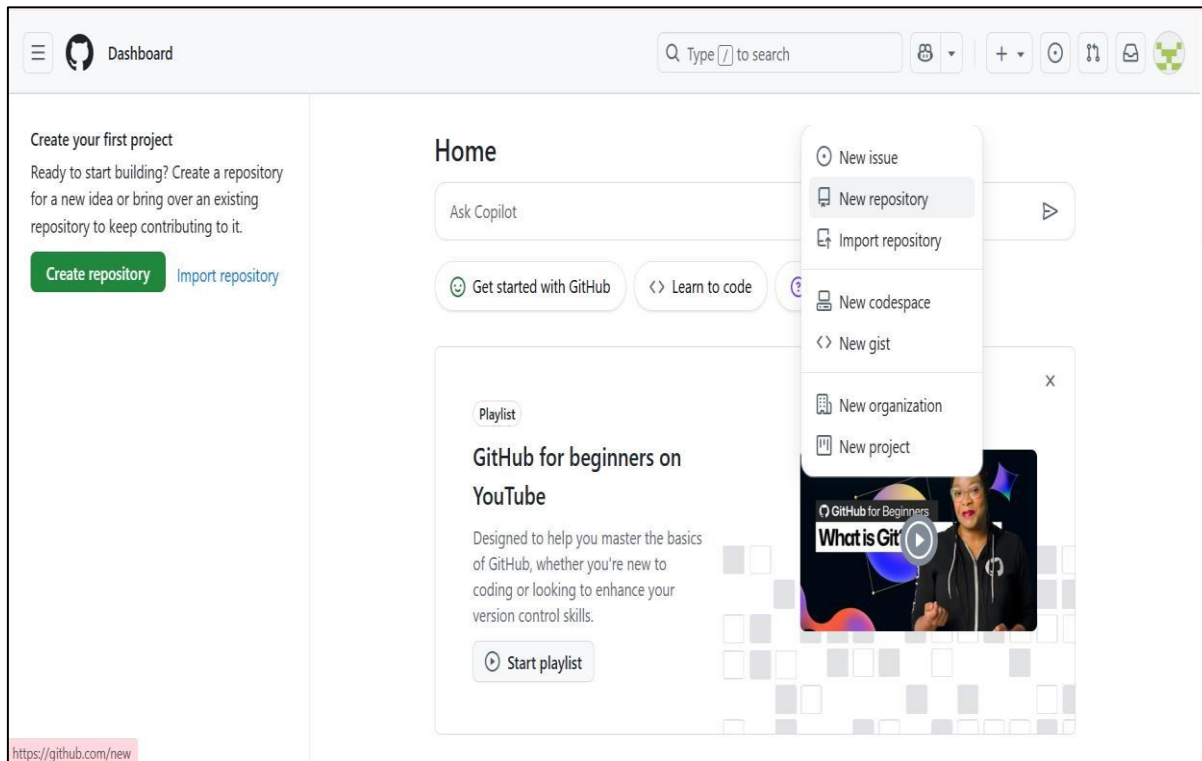
Create account >

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.

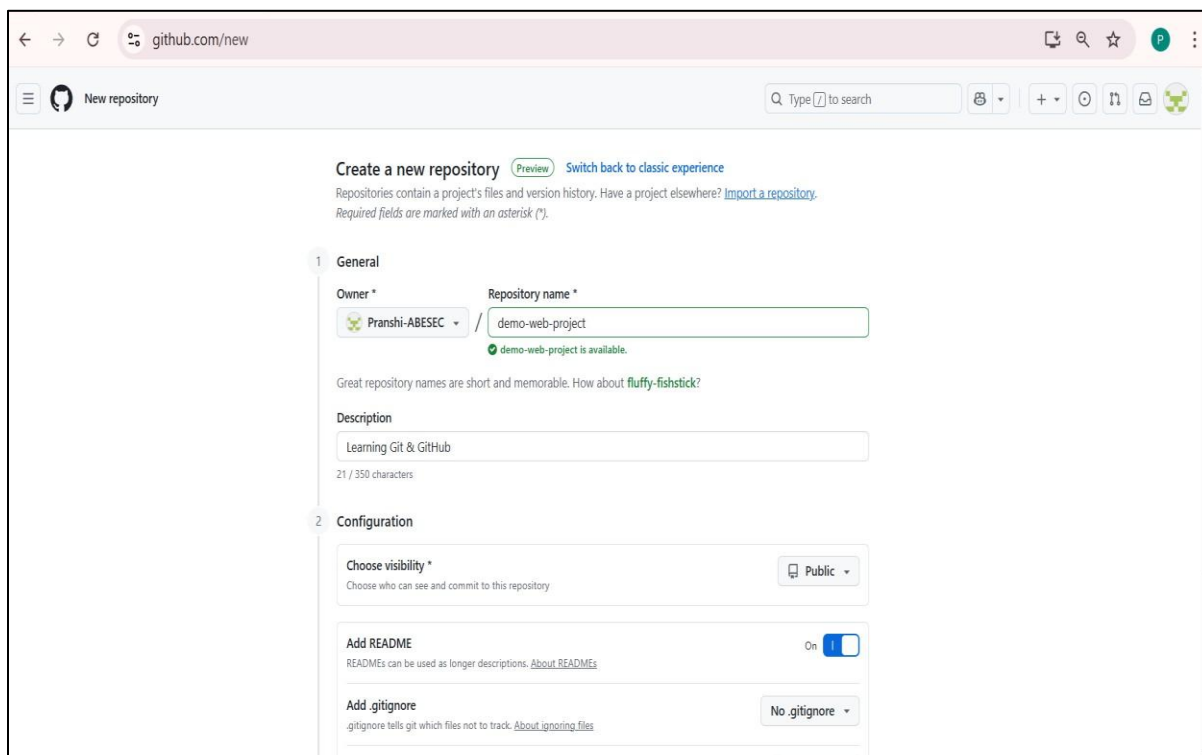


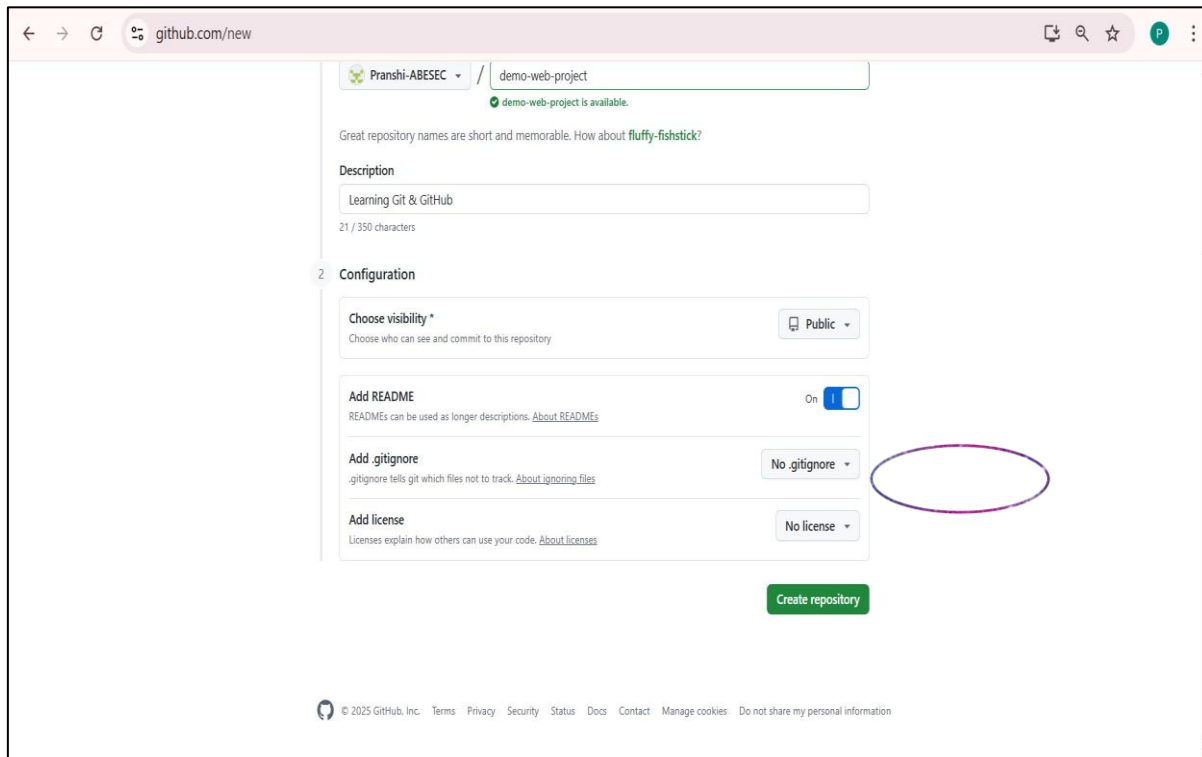
3. Create a new repository:





- Repository Name: demo-web-project
- Description: "Learning Git & GitHub"
- Public → Add README.md → Create Repository.





Creating a Local Repository

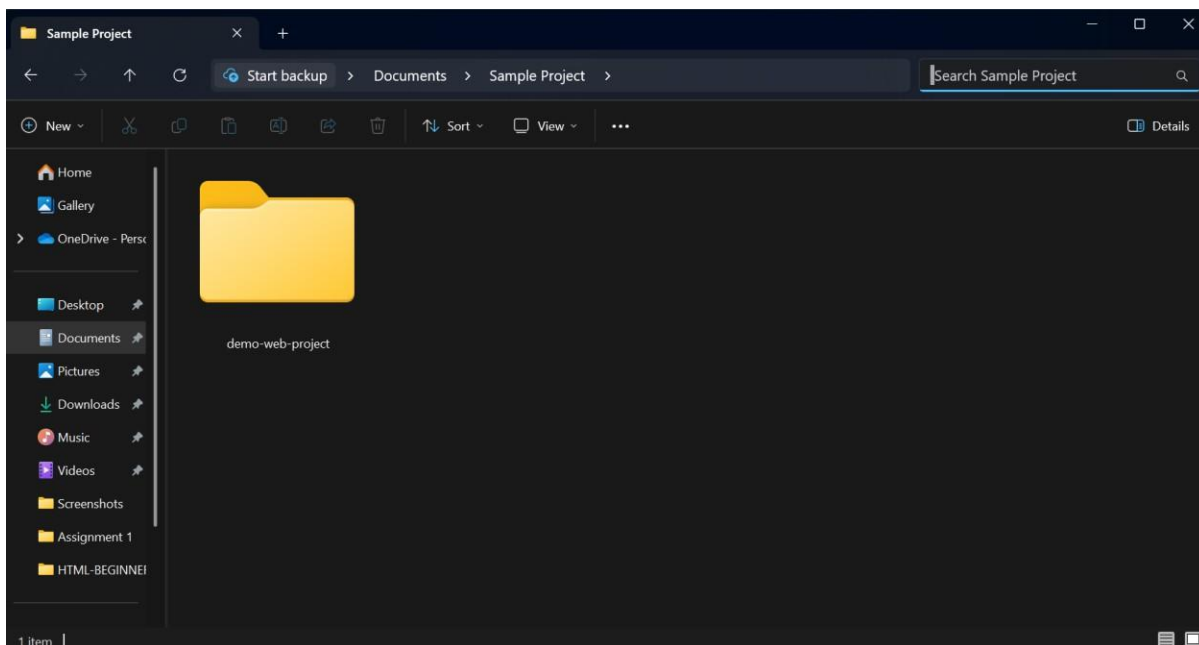
A **local repository** is a project folder on your computer that Git will track. This is the **first step** before pushing anything to GitHub.

You can create it in two ways:

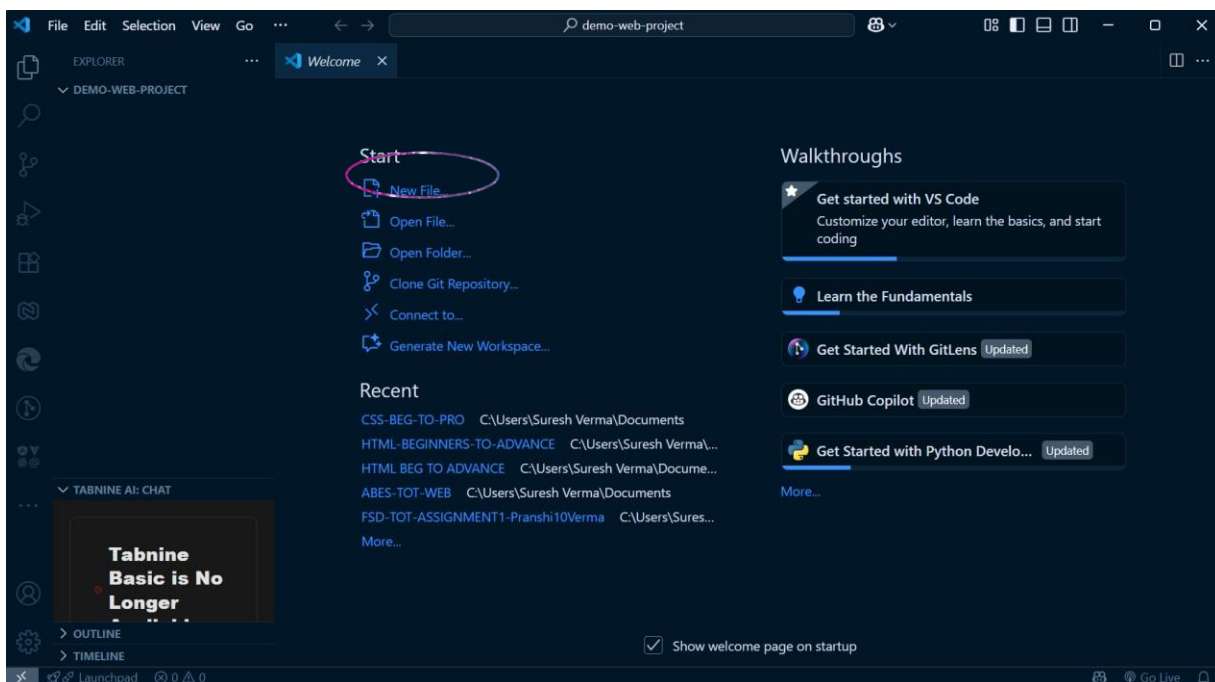
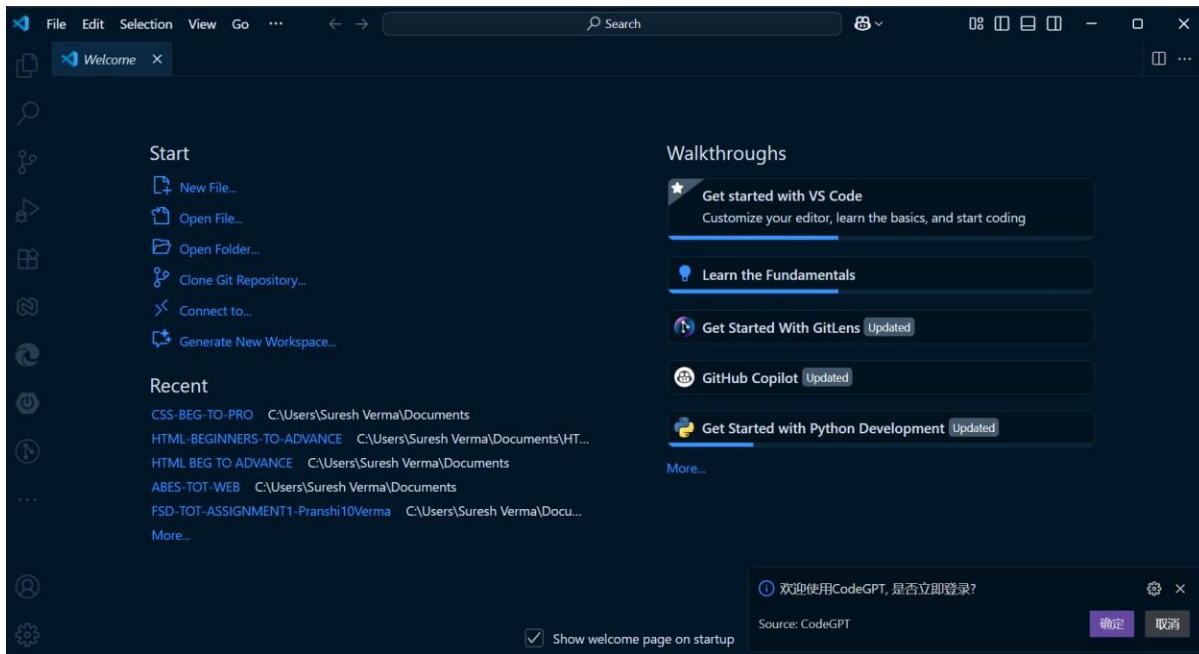
Method A – Using VS Code (Beginner-Friendly UI)

1. Create a project folder

- Example: demo-web-project



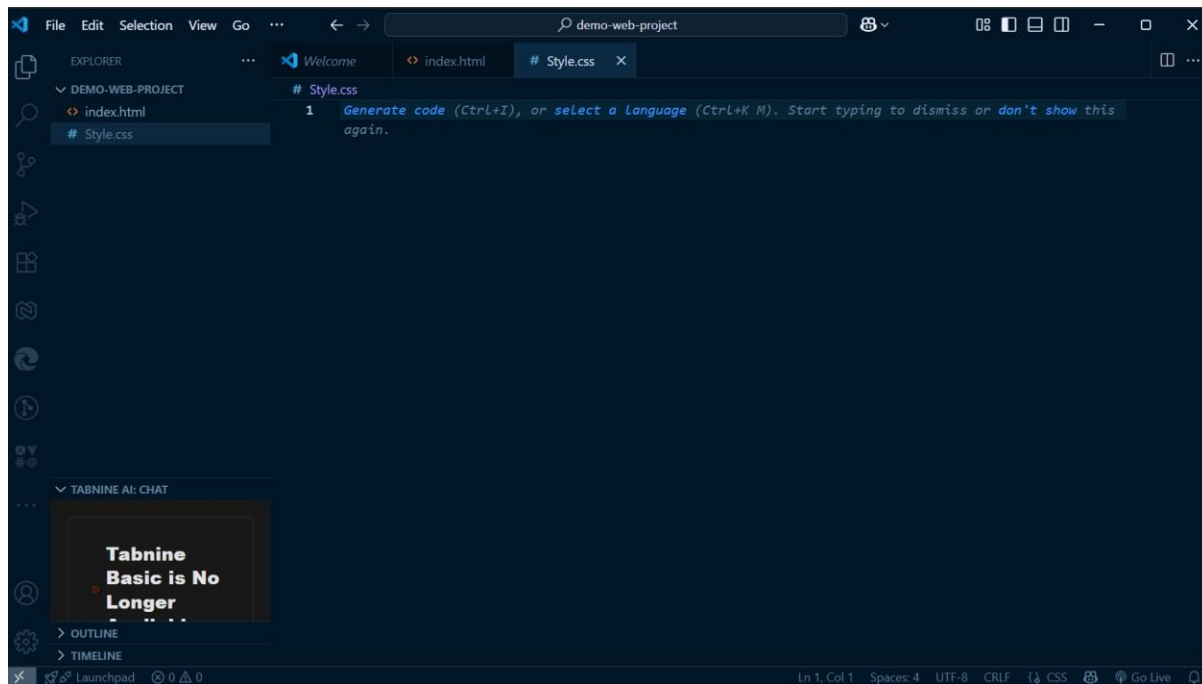
- Open it in VS Code (**File** → **Open Folder**).



2. Create project files

Example:

- index.html
- style.css



index.html

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
<meta charset="UTF-8" />
```

```
<title>My First GitHub Project</title>
```

```
</head>
```

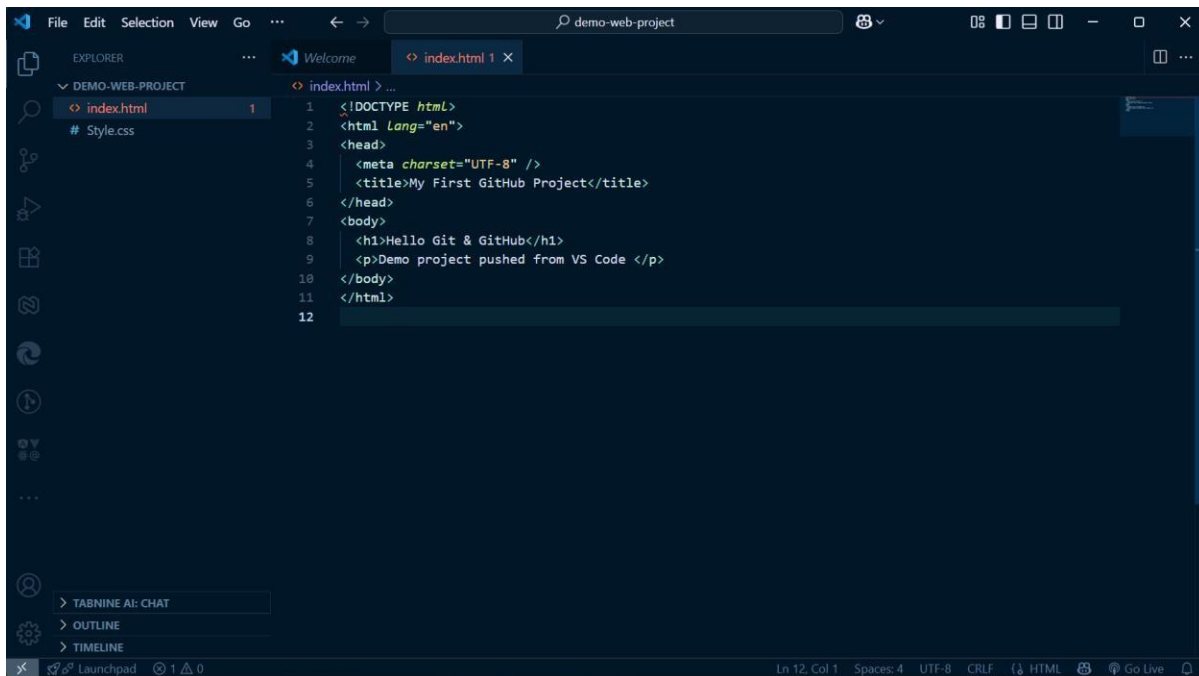
```
<body>
```

```
<h1>Hello Git & GitHub</h1>
```

```
<p>Demo project pushed from VS Code </p>
```

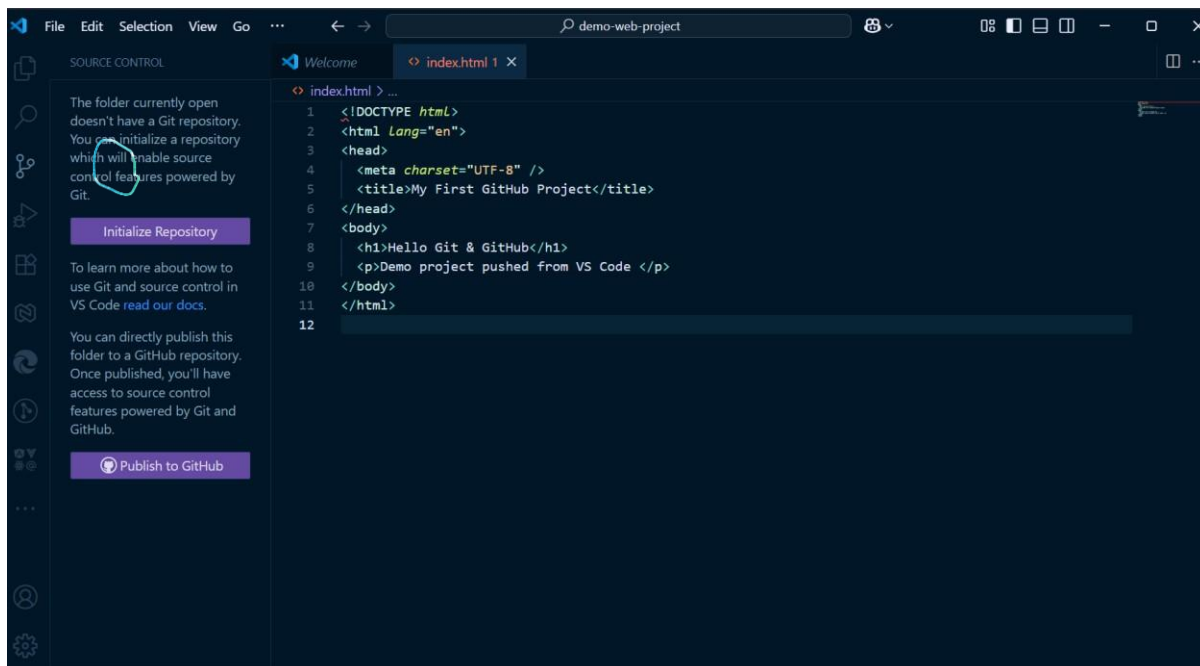
```
</body>
```

```
</html>
```



3. Initialize Git (create local repository)

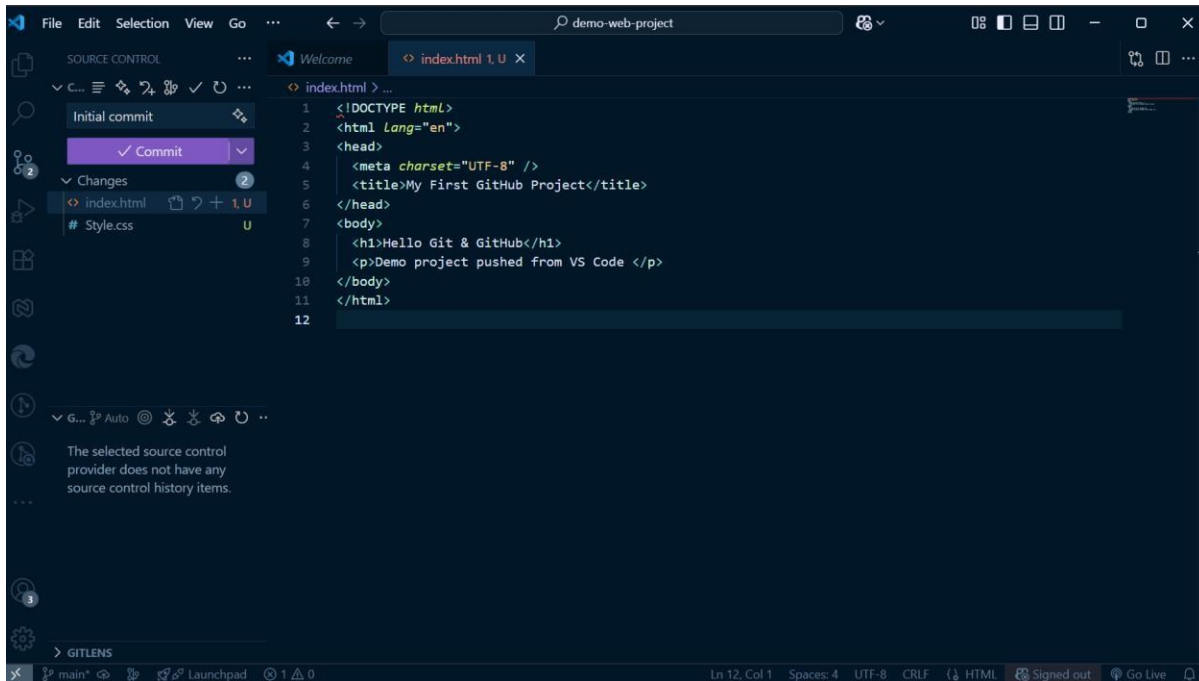
- Go to the **Source Control** panel in VS Code (branch icon on sidebar).



- Click **Initialize Repository**.
- This creates a **local Git repository** inside the folder.

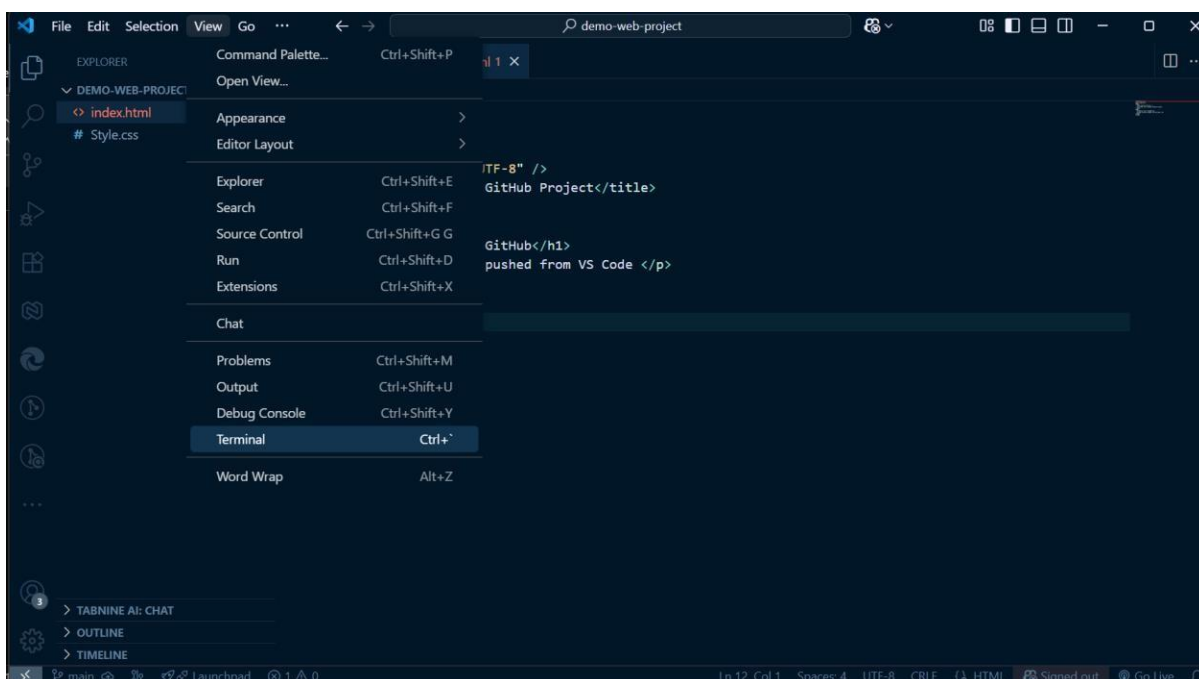
4. Make your first commit

- Stage changes (+ button).
- Enter a commit message: "Initial commit".
- Click **Commit**.



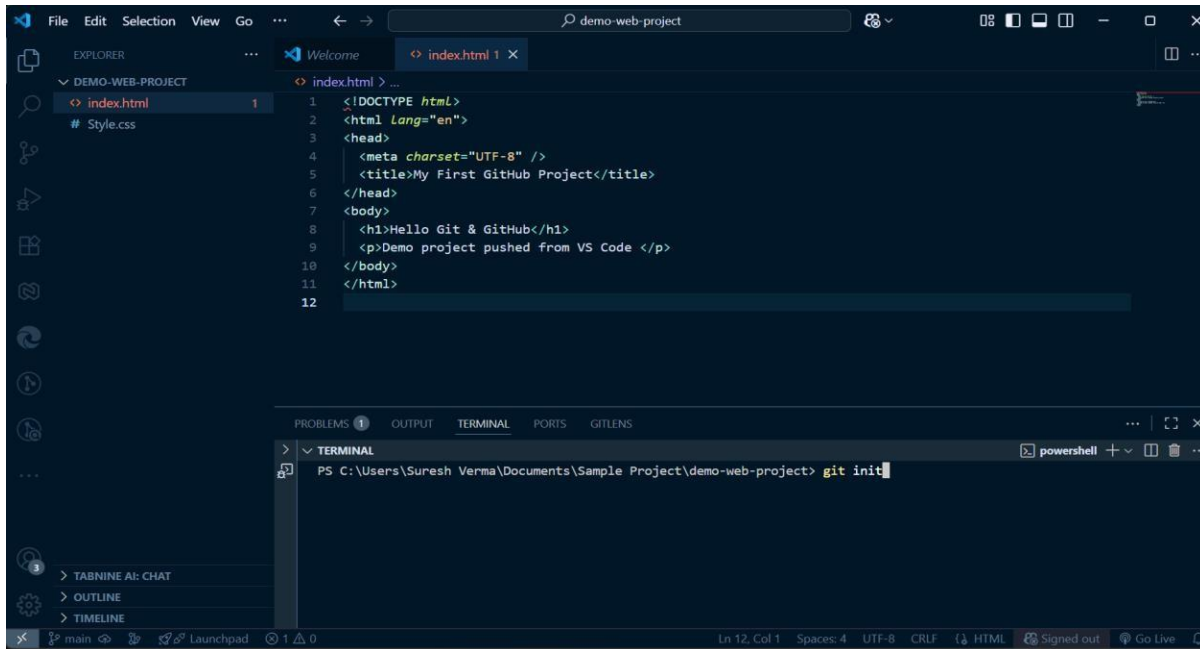
Method B – Using Terminal

1. Open **VS Code Terminal** (Ctrl + ~) or CMD.
2. Navigate to your project folder: `cd path/to/demo-web-project`



Initialize the repository:

git init



The screenshot shows the Visual Studio Code interface with a project named 'demo-web-project'. The Explorer sidebar on the left shows a file named 'index.html' and a folder named 'Style.css'. The main editor window displays the content of 'index.html', which is a basic HTML document with a title 'My First GitHub Project' and a body containing 'Hello Git & GitHub' and 'Demo project pushed from VS Code'. The bottom panel shows the 'TERMINAL' tab with a PowerShell prompt at 'PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project>' and the command 'git init' entered.

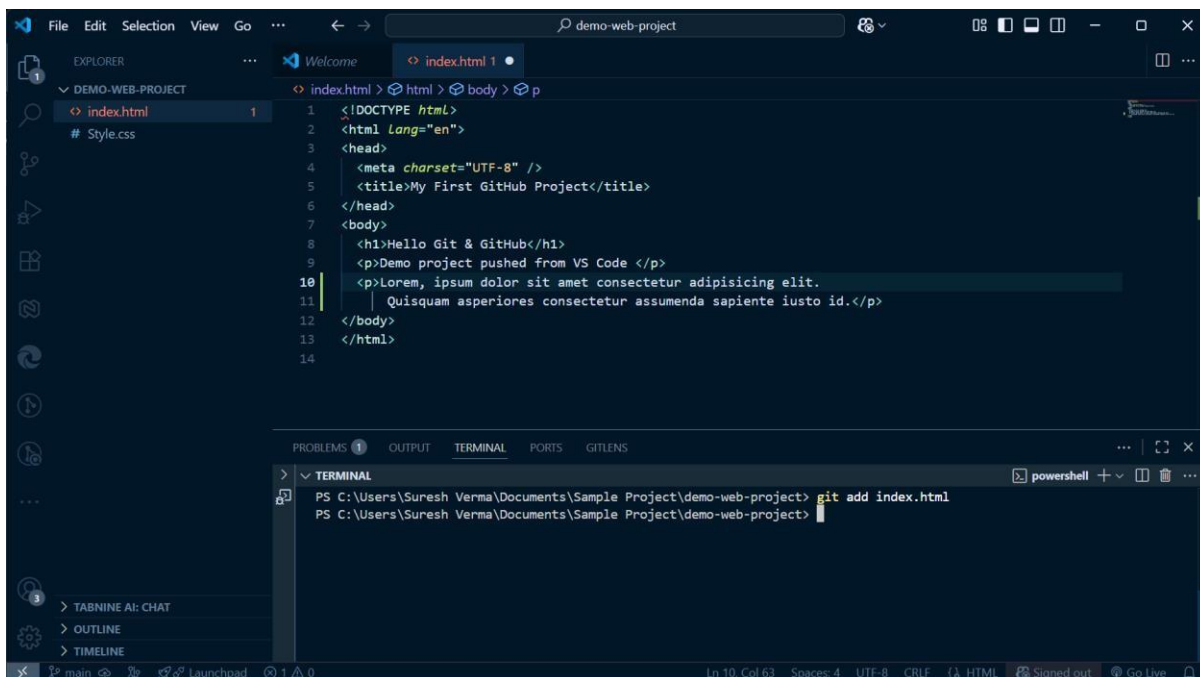
- This command creates a **hidden .git folder**, which means Git is now tracking your project.
- Add and commit files:

Add a Single File

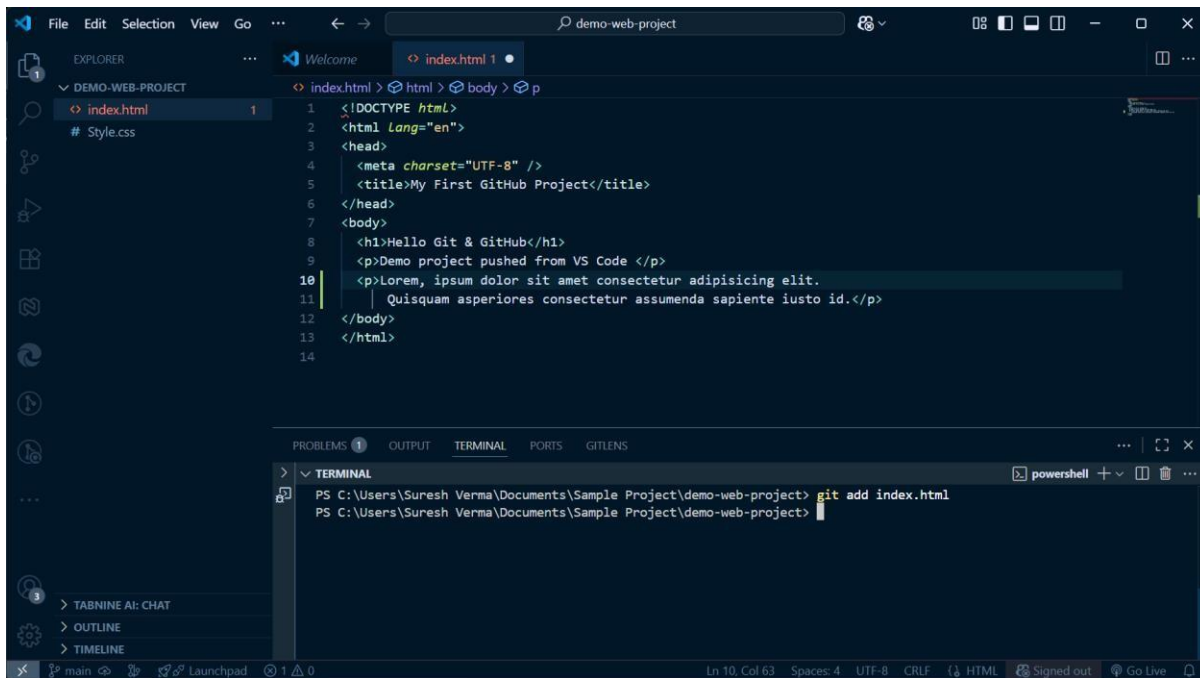
If you only want to add **one file** to the staging area (for example, index.html): *git add index.html*

Add Multiple Specific Files

You can also add more than one file at a time by naming them: *Git add index.html*



The screenshot shows the Visual Studio Code interface with the same 'demo-web-project'. The 'index.html' file in the main editor now contains additional text: 'Lorem, ipsum dolor sit amet consectetur adipisicing elit.' and 'Quisquam asperiores consectetur assumenda sapiente iusto id.' The bottom panel shows the 'TERMINAL' tab with a PowerShell prompt at 'PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project>' and the command 'git add index.html' entered.

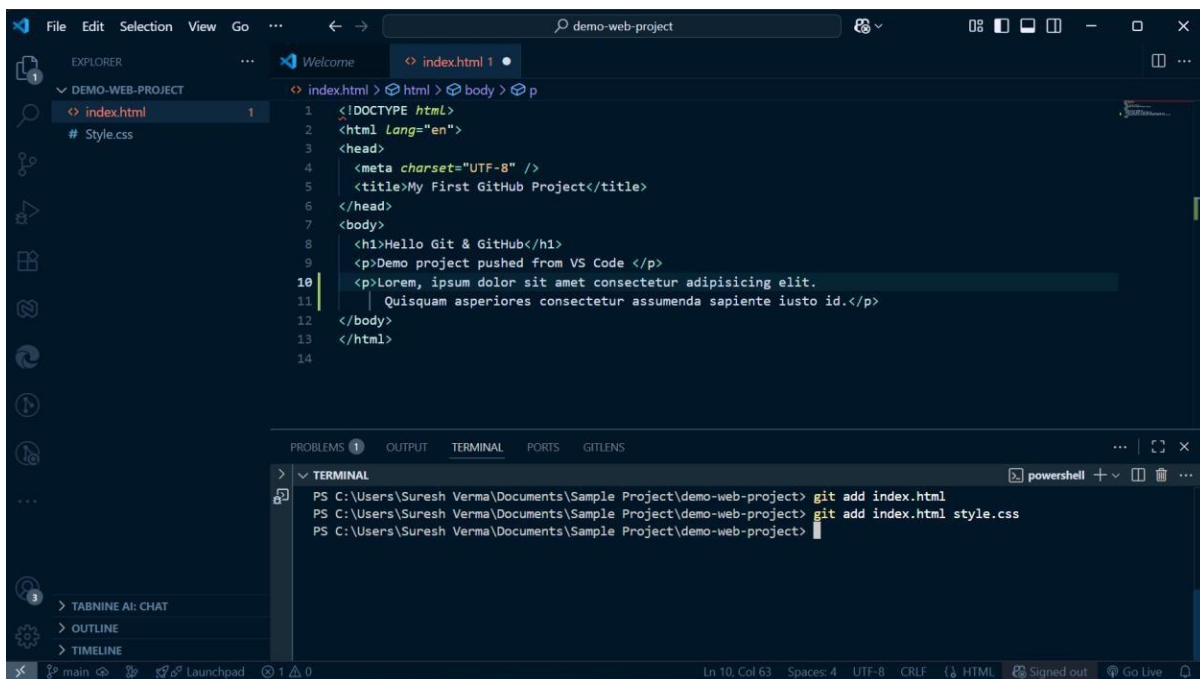


The screenshot shows the Visual Studio Code editor with a file named `index.html` open. The file content is as follows:

```
1 <!DOCTYPE html>
2 <html Lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <title>My First GitHub Project</title>
6 </head>
7 <body>
8   <h1>Hello Git & GitHub</h1>
9   <p>Demo project pushed from VS Code </p>
10  <p>Lorem, ipsum dolor sit amet consectetur adipisicing elit.
11    Quisquam asperiores consectetur assumenda sapiente iusto id.</p>
12 </body>
13 </html>
14
```

The terminal at the bottom shows the command `git add index.html` being executed in a PowerShell window.

git add index.html style.css



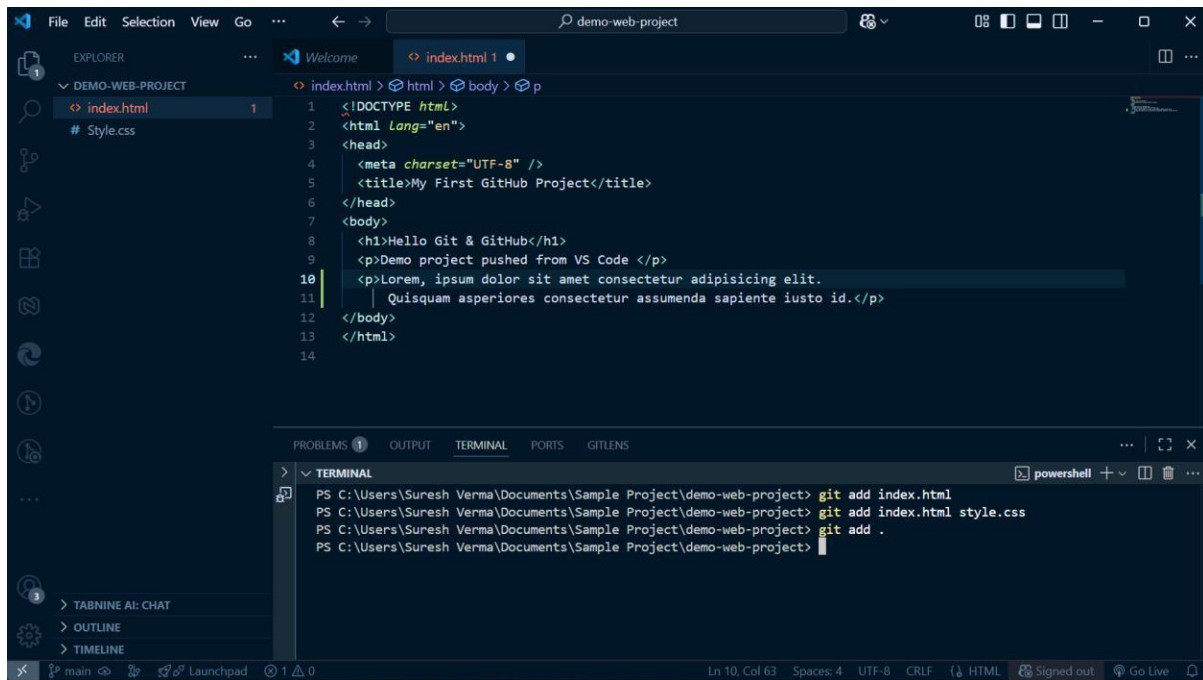
The screenshot shows the Visual Studio Code editor with the same `index.html` file. The terminal at the bottom now shows two commands being executed:

```
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project> git add index.html
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project> git add index.html style.css
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project>
```

Add All Files (Recommended when you want everything)

If you want to add all files in the project folder:

git add .



The screenshot shows the Visual Studio Code interface. The Explorer sidebar on the left shows a project named 'DEMO-WEB-PROJECT' with two files: 'index.html' and 'style.css'. The main editor area displays the content of 'index.html', which is an HTML document with a title 'My First GitHub Project' and some placeholder text. The bottom panel shows the 'TERMINAL' tab with a PowerShell prompt. The terminal contains the following commands and output:

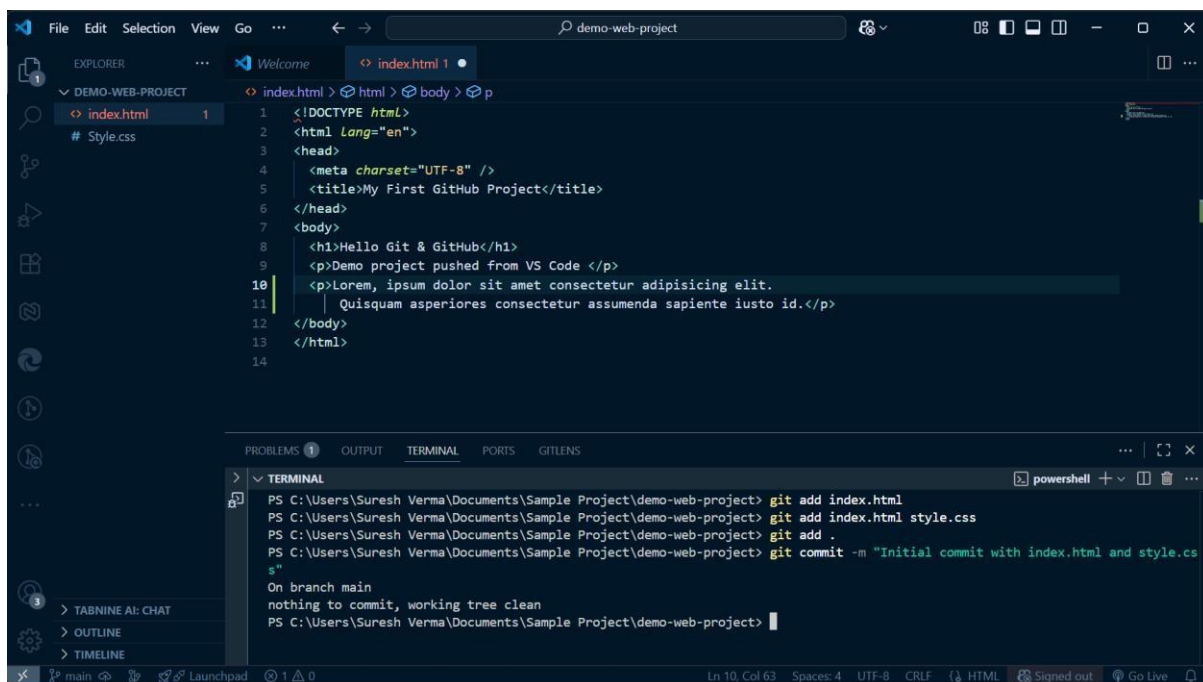
```
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project> git add index.html
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project> git add index.html style.css
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project> git add .
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project>
```

Commit the Changes

After adding files, you need to **commit** them

with a message: `git commit -m "Initial`

`commit with index.html and style.css"`



This screenshot shows the same VS Code interface as the previous one, but the terminal now shows the completion of the commit command. The terminal output is as follows:

```
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project> git add index.html
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project> git add index.html style.css
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project> git add .
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project> git commit -m "Initial commit with index.html and style.css"
[main 1a1b2c3] Initial commit with index.html and style.css
1 file changed, 14 insertions(+), 0 deletions(-)
create mode 100644 index.html
PS C:\Users\Suresh Verma\Documents\Sample Project\demo-web-project>
```

A **commit** is like saving a snapshot of your project at that

point in time. Now we will link our local repository with

the GitHub repository (remote). **Now your project is on GitHub!**